Day 119 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Model a Geometry Design

Standards: 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can model a Geometry Design, and explain and check my reasoning.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 4, 19, 23

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01 / Learn

Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

Plot and connect (0,0), (7,0), (7,8), (0,8). Find the side lengths and area.

1. Plot vertices in order and connect them, including the last to the first.

2. Equal x-values make a vertical side; equal y-values make a horizontal side.

3. Horizontal length 7 - (0) = 7

4. vertical length 8 - (0) = 8

5. area = 7 × 8.

6. State the result in the form requested: Sides 7 and 8; area 56 square units.

Answer: Sides 7 and 8; area 56 square units

Worked example 2

A prism measures 13/2 m by 5/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 13/2 × 5/2 × 1 = 65/4.

3. The edge counts are 13, 5, 2, giving 130 cubes.

4. Each cube is 1/8 m^3

5. 130/8 = 65/4.

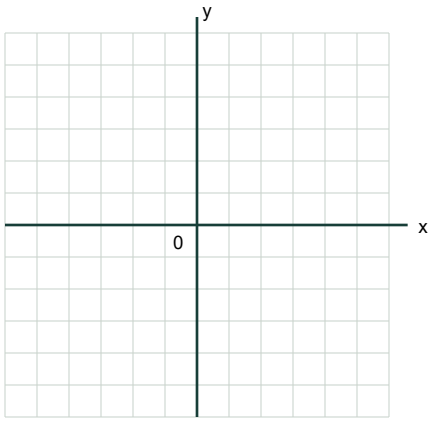
Answer: 65/4 m^3; 130 cubes

Day 119 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Plot and connect (0,0), (11,0), (11,8), (0,8). Find the side lengths and area.



Label the scale and units.

02 / Constructed response

An L-shaped garden is a 9 m by 10 m rectangle with an 6 m by 7 m corner rectangle removed. Sketch and find the garden area.

03 / Constructed response

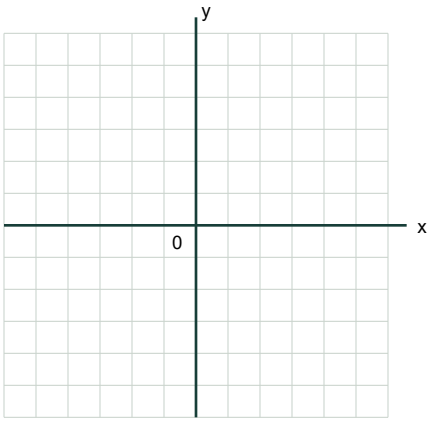
A 8 cm by 5 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A rectangular prism has edges 2, 1, and 2 cm. Sketch it and find its volume.

05 / Constructed response

Plot and connect (-5,-5), (2,-5), (2,2), (-5,2). Find the side lengths and area.



Label the scale and units.

06 / Constructed response

An L-shaped garden is a 9 m by 9 m rectangle with an 3 m by 3 m corner rectangle removed. Sketch and find the garden area.

07 / Constructed response

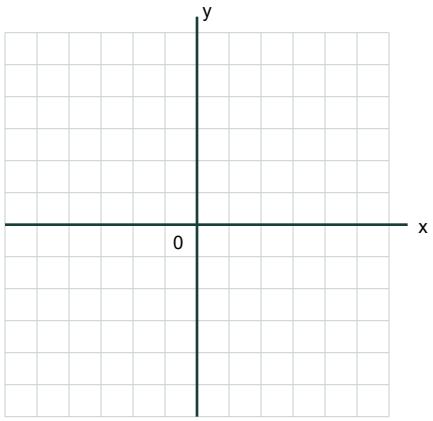
A 2 cm by 3 cm base and 2 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Numeric entry and model reasoning

A prism measures 7/2 m by 3/2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Coordinate plotting and two-part response

A rectangle has vertices (-7, -4), (4, -4), (4, 4), and (-7, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

10 / Constructed response

A 11 m by 9 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 5 m. Sketch the remaining polygon and find its area.

Day 119 | Unit 08 | Answer key

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Learning target: I can model a Geometry Design, and explain and check my reasoning.

1. Sides 11 and 8; area 88 square units

Horizontal length 11 - (0) = 11; vertical length 8 - (0) = 8; area = 11 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

2. 48 m²

Outer area 90 minus cutout 42 = 48.

6.G.A.1 | Constructed response

3. 158 cm²

Closed surface area = 2(40+24+15) = 158.

6.G.A.4 | Constructed response

4. 4 cm³

B = 2 × 1 = 2 cm²; V = B × 2 = 4 cm³.

6.G.A.2 | Constructed response

5. Sides 7 and 7; area 49 square units

Horizontal length 2 - (-5) = 7; vertical length 2 - (-5) = 7; area = 7 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

6. 72 m²

Outer area 81 minus cutout 9 = 72.

6.G.A.1 | Constructed response

7. 26 cm²

Closed surface area = 2(6+4+6) = 32. Remove the top area 6: 26.

6.G.A.4 | Constructed response

8. 63/8 m³; 63 cubes

Multiply edges: 7/2 × 3/2 × 3/2 = 63/8. The edge counts are 7, 3, 3, giving 63 cubes. Each cube is 1/8 m³; 63/8 = 63/8.

6.G.A.2 | Numeric entry and model reasoning

9. 11 by 8 units; area 88 square units

Horizontal length = 7 + 4; vertical length = 4 + 4. Multiply 11 × 8 = 88.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

10. 89 m²

Whole rectangle 99; remove triangle 10; difference 89.

6.G.A.1 | Constructed response